

CORE JAVA NOTES

Overview of the topics:

1. Java Introduction & Architecture

- What is Java
- Features of Java
- JDK, JRE, JVM
- Java Compilation Flow
- Stack vs Heap Memory

2. Java Basics & Fundamentals

- Variables & Data Types
- Type Casting
- Operators
 - Arithmetic Operators
 - Relational Operators
 - Logical Operators
 - Bit Manipulation Operators
 - `&`
 - `|`
 - `^`
 - `~`
 - `<<`
 - `>>`
- Input/Output
- Conditional Statements
- Loops
- Methods
- Recursion
- Math Class Methods
 - `Math.max()`
 - `Math.min()`
 - `Math.abs()`
 - `Math.pow()`

3. Object-Oriented Programming (OOP)

- Classes & Objects
- Instance Variables
- Methods
- Constructors
- Object References
- `this` keyword

- `static` keyword
- Encapsulation (basic)
- Inheritance (basic)
- Polymorphism (basic)
- Abstraction (basic)
- Access Modifiers
- Method Overloading
- Method Overriding
- `extends` and `super`

4. Arrays, Strings & Memory Sequences

- 1D Arrays
- 2D Arrays
- String Class
- String Methods
- String Immutability
- StringBuilder
- Character Utility Methods
 - `Character.isDigit()`
 - `Character.isLetter()`
 - `Character.toLowerCase()`
 - `Character.toUpperCase()`

5. Generics & Wrapper Classes

- Wrapper Classes
- Autoboxing & Unboxing
- Generic Syntax for Collections

6. Java Collection Framework

- ArrayList
- LinkedList
- HashSet
- HashMap
- Queue
- ArrayDeque
- PriorityQueue (Heap)
- Nested Collections
 - `List<List<Integer>>`

7. Iteration Techniques

- for-each Loop
- Iterator
- Traversing List

- Traversing Set
- Traversing Map

8. Sorting & Comparators

- `Arrays.sort()`
- `Collections.sort()`
- Comparable
- Comparator
- Custom Sorting
- Lambda Sorting

9. Exception Handling

- try-catch
- finally
- throw & throws

10. Important Java Interview Concepts

- `==` vs `.equals()`
- `final` keyword
- `static` keyword
- Pass by Value
- Immutable Strings
- Garbage Collection (basic)
- HashMap Internal Working (basic)
- Time Complexity of Collections

11. High-Level Understanding Only

- File Handling
- JDBC
- Reflection
- Serialization
- Advanced Multithreading
- Design Patterns
- Networking
- Advanced Generics
- JavaFX/Swing